

# ***Introduction to Computer Architecture***

## **Sheet #9**

### **Exercises:**

1. Suppose a computer using direct mapped cache has  $2^{20}$  words of main memory, and a cache of 32 blocks, where each cache block contains 16 words.
  - a. How many blocks of main memory are there?
  - b. What is the format of a memory address as seen by the cache, i.e., what are the sizes of the tag, block, and word fields?
  - c. To which cache block will the memory reference  $0DB63_{16}$  map?

### **ANSWER:**

- a.  $2^{20}/2^4 = 2^{16}$
  - b. **20-bit addresses with 11 bits in the tag field, 5 in the block field, and 4 in the word field.**
  - c.  **$0DB63 = 00001100101\ 10110\ 0111$ , which implies Block 22**
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2. Suppose a computer using direct mapped cache has  $2^{32}$  words of main memory, and a cache of 1024 blocks, where each cache block contains 32 words.
  - a. How many blocks of main memory are there?
  - b. b. What is the format of a memory address as seen by the cache, i.e., what are the sizes of the tag, block, and word fields?
  - c. c. To which cache block will the memory reference  $000063FA_{16}$  map?

### **ANSWER:**

- a.  $2^{32}/2^5 = 2^{27}$
  - b. **32-bit addresses with 17 bits in the tag field, 10 in the block field, and 5 in the word field.**
  - c.  **$000063FA = 000000000000000000\ 1100011111\ 11010$ , which implies Block 799**
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3. Suppose a computer using fully associative cache has  $2^{16}$  words of main memory, and a cache of 64 blocks, where each cache block contains 32 words.
  - a. How many blocks of main memory are there?
  - b. What is the format of a memory address as seen by the cache, i.e., what are the sizes of the tag and word fields?

- c. To which cache block will the memory reference F8C9 map?

**ANSWER:**

- a.  $2^{16}/2^5 = 2^{11}$
  - b. **16-bit addresses with 11 bits in the tag field and 5 in the word field**
  - c. **Since it's associative cache, it can map anywhere**
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4. Suppose a computer using fully associative cache has  $2^{24}$  words of main memory, and a cache of 128 blocks, where each cache block contains 64 words.
- a. How many blocks of main memory are there?
  - b. What is the format of a memory address as seen by the cache, i.e., what are the sizes of the tag and word fields?
  - c. To which cache block will the memory reference 01D87216 map

**ANSWER:**

- a.  $2^{24}/2^6 = 2^{18}$
  - b. **24-bit addresses with 18 bits in the tag field and 6 in the word field**
  - c. **c. Since it's associative cache, it can map anywhere**
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5. Assume a system's memory has 128M words. Blocks are 64 words in length and the cache consists of 32K blocks. Show the format for a main memory address assuming a 2-way set associative cache mapping scheme. Be sure to include the fields as well as their sizes.

**ANSWER:**

**Each address has 27 bits, and there are 7 in the tag field, 14 in the set field and 6 in the word field.**

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6. A 2-way set-associative cache consists of four sets. Main memory contains 2K blocks of eight words each.
- a. Show the main memory address format that allows us to map addresses from main memory to cache. Be sure to include the fields as well as their sizes.
  - b. Compute the hit ratio for a program that loops 3 times from locations  $8_{10}$  to  $51_{10}$  in main memory. You may leave the hit ratio in terms of a fraction.

**ANSWER:**

- a.  $2K * 2^3 = 2^{14}$ , so we have 14-bit addresses with 9 bits in the tag field, 2 bits in the set field (since we have four sets), and 3 in the word field
  - b. First iteration of the loop: Address 8 is a miss, then entire block brought into Set 1. 9-15 are then hits. 16 is a miss, entire block brought into Set 2, 17-23 are hits. 24 is a miss, entire block brought into Set 3, 25-31 are hits. 32 is a miss, entire block brought into Set 0, 33-39 are then hits. 40 is a miss, entire block brought into Set 1 (note we do NOT have to throw out the block with address 8 as this is 2-way set associative), 41-47 are hits. 48 is a miss, entire block brought into Set 2, 49-51 are hits. For the first iteration of the loop, we have 6 misses, and  $5*7 + 3$  hits, or 38 hits. On the remaining iterations, we have  $5*8+4$  hits, or 44 hits each, for 88 more hits. Therefore, we have 6 misses and 126 hits, for a hit ratio of 126/132, or 95.45%.
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- 7. Suppose a computer using set associative cache has  $2^{16}$  words of main memory, a cache of 32 blocks and each cache block contains 8 words.
  - a. If this cache is 2-way set associative, what is the format of a memory address as seen by the cache, i.e., what are the sizes of the tag, set, and word fields?
  - b. If this cache is 4-way set associative, what is the format of a memory address as seen by the cache?

**ANSWER:**

- a.  $2^{16}$  words of main memory implies we have 16 bits in an address. Cache contains  $2^5$  blocks, but each set must have 2 blocks, so we have  $2^5/2=2^4$  sets. Therefore our 16-bit address is divided into 9 bits for the tag field, 4 bits for the set field, and 3 bits for the word field.
- b. The 32 blocks in cache must now be divided into sets with 4 blocks each, implying we have only 8 sets. The tag field would now have 10 bits, the set field 3 bits, and the word field 3 bits.